

Qualifyze

AI Data Scientist Case Study

Status

Final (July 2025)

Scope

This document outlines a practical business case to complement the interview process

Context

Qualifyze aims to become a trust layer for global supply chain risk management, relying on insights derived from diverse datasets, including audit reports, supplier-provided data, public sources, and regulatory documents such as **non-compliance reports** and **warning letters**. A Data AI Scientist plays a key role in developing predictive models and AI-driven solutions to enhance the accuracy, consistency, and prediction using these data.

Your challenge is to analyze, preprocess, and model data to provide meaningful insights and automation that support ****supply chain risk assessment and supplier verification****, particularly leveraging ****historical non-compliance reports and warning letters****.

For most questions, “it depends” will likely be the answer. Still, we want to understand your perspective on things, your hypotheses and your decision-making process and the evaluation criteria you apply.

If you are missing data, make assumptions, explain why, and continue. Should you feel you need more information or context to answer a specific section, please feel free to reach out at any time to obtain it in preparation for the case study discussion.

Supporting Materials

- **Website:** <https://www.qualifyze.com>
- **Data source:** [Warning letters](#)
- **Information:** [Inspection Classifications](#)
- **Data Source:** [broader FDA data-sets](#)
- **Data Source:** [FDA Dashboards](#)

Format & Preparation

You will present your approach to selected members of the Qualifyze product and engineering teams. Any material (notebooks, reports, slides) should be shared 24 hours before the meeting.

The discussion will be around your methodology, assumptions, trade-offs, and recommendations.

Feel free to ask questions before making assumptions. Clear communication is crucial to success in this role.

Business Challenge

Qualifyze collects structured and unstructured data from multiple sources to evaluate suppliers. However, several challenges arise:

- Data inconsistency across sources (e.g., discrepancies in supplier names, locations, certifications).
- Missing or incomplete information.
- The need for predictive analytics to assess supplier risks based on historical non-compliance reports and warning letters.
- The necessity to interpret AI models for regulatory compliance.

Task

Your objective is to:

1. Data Processing & Feature Engineering (High level)
 - a. Define a strategy to import and standardize non-compliance reports and warning letters from different sources.
 - b. Link Public and Private Data.
 - c. Propose a data preprocessing pipeline to clean, normalize, and enrich supplier-related data.
 - d. Handle missing values and inconsistent data across multiple sources.
2. Predictive Modeling & Insights (Prototype)
 - a. Develop a prototype of a predictive model to assess the likelihood of future non-compliance for a given supplier based on historical audit data, and warning letters.
 - b. Define key features influencing supplier risk.
 - c. Evaluate model performance and discuss trade-offs.
3. Interpretability

- a. Ensure the model is explainable.
 - b. Discuss how to make AI decisions transparent to stakeholders.
- 4. Scalability & Deployment (Nice to Have)
 - a. Provide recommendations on deploying the model in a production environment.
 - b. Suggest improvements for efficiency and accuracy at scale.

Expected Deliverables

- A github repository with the solution
- A report summarizing key findings, methodologies, and recommendations.
- A short presentation for discussion.